Supplemental Material

Table S1. Basic characteristics of the included studies.

Study	Study design	Patients included	Age	Prior stroke/TIA n (%)	Recorder type	Indication for device implantation	Anticoagulation n (%)	AHRE definition	Time cut offs	Follow-up (months)	Outcome	Study population
Glotzer et al 2003	Randomized	312	74	55 (17,6)	PMs	sinus node dysfunction	NA	longest episode	5 min	27	Stroke	MOST
Capucci et al 2005	Observational Prospective	725	71 ±11	13 (1,8)	PMs	class I or II ACC/AHA indication for dual-chamber pacing: node disease (82.8%) AV block (4.7%), drug-induced bradycardia in (4.4%), other (8.1%)	261(36,4)	longest episode	24 h	22 (16-30)	Stroke	NA
Botto et al 2008	Observational,retrospective	568	70 ± 10	8 (1,4)	PMs	bradycardia according to current guidelines	165 (29)	longest episode	>5 min >24 h	12	Stroke/ TIA	NA
Caldwell et al 2009	Observational, retrospective	162	66 ±1.8	NA	CRT	HF NYHA class III –IV CHF	NA	longest episode	30 sec	14.1 ±1	Stroke or SE	NA
Bertini et al 2010	Observational,prospective	495	62.2 ±11.7	NA	ICD	heart failure according to the current AHA/ACC/ESC guidelines	263(54)	longest episode	10 min	16.4 ±11.2	Stroke or SE	NA
Shanmugan et al 2011	Observational,prospective	560	66 ± 10	1 (0.18)	CRT	HF with no history of AF	67 (12.0)	day-level cumulative burden	14 min (3.8 h as threshold)	12.3 (IQR 8.4-13)	Stroke or SE	EVEREST
Jons et al 2011	Randomized	271	63,3 ± 11	NA	ICM	post MI ≤40% EF	56 (20,6)	longest episode	30 sec	24	Stroke	CARISMA
Healey et al 2012	Prospective Observational	2580	76 ± 7	312 (12,1)	PMs or ICD	sinus-node or atrioventricular-node disease or any indication for ICD	0(0)	longest episode	6 min	30	Stroke or SE	ASSERT*
Petrac et al, 2012	Observational, retrospective	308	67 ± 10	NA	PMs	second- or third-degree AV block	48 (15,6)	longest episode	5min	36 ± 20	Stroke	NA
Gonzalez et al 2014	Observational,retrospective	224	74 ± 12	13 (6)	PMs	Sinus node dysfunction AV block	7 (3.1)	longest episode	5 min	79.2 ± 24	Stroke	NA
Christensen et al 2014	Observational,prospective	85	56,7	85 (100)	ILR	cryptogenic stroke	18 (20,7)	longest episode	2 min	19 ±10.3	Stroke	SURPRISE
Boriani et al 2014	Observational, pooled analysis from five prospective studies	10016	70	589(6)	CIEDs	class I/II indication for an implantable cardiac rhythm device	1822(18)	day-level cumulative burden	5 min 1 h 6 h 12 h 23 h	24 (14–40)	Stroke	TRENDS, PANORAMA
Witt et al 2015	Observational, retrospective	394	67 (59-74)	58 (14,7)	CRT	HF (standard indication for CRT treatment)	56(14,2)	longest episode	6 min	50.4 (IQR 30–79.2)	Stroke or SE	NA
Kim et al 2016	Observational,retrospective	880	62,7 ± 14	70 (8)	PM, ICDs, and CRTs	classes I-II recommendation of the current ACCF/AHA/HRS guidelines for device implantation	40 (4,5)	longest episode	5 min	55 (20-90.2)	Stroke	NA
Wilton et al 2016	Randomized, prospective	972	66.1	NA	ICDs and CRTs	HF	286 (29,4)	longest episode	30 sec	41± 19	Stroke	RAFT
Swiryn et al, 2016	Observational,	5379	69,7	186 (5.9%)	PMs and ICDs	indication for a cardiac rhythm management	823 (15,3)	longest episode	>20 sec	22,9	Stroke or SE	RATE

	prospective					implantable device						
Reiffel et al, 2017	Observational, prospective	326	71,5 ± 9.9	80 (20.3)	ICM (Reveal XT or Reveal LINQ; Medtronic) .	high risk patients for AF	72 (56,3)	longest episode	6 min	22,5	Stroke	REVEAL AF
Israel et al 2017	Observational, prospective	123	65 ± 9	123 (100)	ILR	ESUS	NA	longest episode	2 min	12.7 ±5.5	Stroke	NA
Amara et al 2017	randomized, single-blind	595	79 ± 8	60 (10,1)	PMs	Sinus node dysfunction, AV block and other conduction defects	0(0)	day-level cumulative burden	6 h	12.8 ± 3.3	Stroke	SETAM
Kawakamy et al 2017	Observational, retrospective	343	80±7	52 (15)	PMs	sinus node disease or atrioventricular block	53 (15)	longest episode	6 min	52± 30	Stroke or SE	NA
Martin et al 2017	Randomized	2718	64,4	243(8,9)	ICD and CRTs	Current clinical class I or II indications for ICD / CRT implantation	302 (11)	longest episode	10 sec	24	Stroke or SE	IMPACT
VanGelder et al 2017	Observational, prospective	2455	76,3 ± 6,7	297 (12)	PMs and ICD	PCM for sinus node or AV node disease, ICD for any indication	0	longest episode	6 h 24h	30	Stroke or SE	ASSERT*
Nakano et al 2018	Observational, retrospective	348	70±16	NA	PMs ICDs, and CRTs	Class I or IIa indication according to the Japanese Circulation Society	0(0)	longest episode	30 sec	65±58	Stroke	NA
Pedersen et al 2018	Observational, prospective	105	65.4 (27.2 - 0.8)	105 (100)	ILR	TIA patients . CHA_2DS_2-Vasc 4	0(0)	longest episode	2 min	12.7 (12.4-13)	Stroke	NA
Perino et al 2019	Observational, retrospective	10212	72±10	0 (0)	CIEDs	Database of CIEDs (not mentioned)	1032 (10)	day-level cumulative burden	6 min 1h 6 h 24 h	45	Stroke	Veterans Affair National Patient Care Database
Li et al 2019	Observational, prospective	594	69 ± 14	59 (9.9)	PMs, ICDs, CRTs	NA	NA	longest episode	5 min	50.4	Stroke or SE	NA
Kaplan et al 2019	Observational, retrospective	21768	68.6±12.7	3047 (14)	IPMs, ICDs, CRTs	according to ACC/AHA guidelines	0(0)	longest episode	6 min and 23.5 h	6 months	Stroke or SE	Optum© Electronic Health Record database, Medtronic CareLinkTM database of CIEDs
Miyazawa et al 2019	Observational, retrospective	856	72.0 (62.0–80.0)	92 (10.7)	ICDs, CRT	current indications for ICD / CRT implantation according to ESC guidelines	151 (19.7)	longest episode	5 min	48.2 ± 32.3	Stroke or SE	NA

CIEDs: cardiac implantable electronic devices, PM: pacemaker, ICD: implantable cardioverter defibrillators, ICM: impantable cardiac monitor, CRT: cardiac resynchronization therapy, ILR: implantable loop recorder, ESUS: embolic stroke of undetermined source, NA: not applicable
***: Both studies conducted in the ASSERT population**

Table S2. Studies providing raw events or incidence rates.

Author	Thresholds	Patients included	Relative Risk (95% CI)		Events	IR (%/yr)
		30 sec				
Caldwell et al 2009	<30sec	74	2,74	0,14 - 52,70	0.86*	1,16
	≥30sec	27			0.86*	3,18
Nakano et al 2018	<30sec	293	6,93	3,20 - 14,90	10	0.7
	≥30sec	55			13	4.3
		2min				
Christensen et al 2014	<2min	69	3,29	1,26 - 8,57	7	6.5
	≥2min	18			6	21.3
Israel et al 2017	<2min	94	1,25	0,49 - 3,20	13	12.8
	≥2min	29			5	15.9
Pedersen et al 2018	<2min	98	1,59	0,23 - 10,81	6	6.1
	≥2min	7			1	14.3
		5min				
Botto et al 2008	<5min	166	2,07	0,46 - 9,30	2	1.2
	≥5min	402			10	2.48
Petrac et al, 2012	<5min	274	1,34	0,31 - 5,75	12	1.46
	≥5min	34			2	1.56
Reiffel et al, 2017	<6min	198	1,55	0,32 - 7,55	3	0.66
	≥6min	128			3	1.56
Kaplan et al 2019	<6min	19443	1,46	1,14 - 1,87	158	NA
	≥6min	8589			102	
		10 min				
Bertini et al 2010	<10min	309	0,92	0,10 - 8,12	4	0.97
	≥10min	84			1	0.89
		6h				
VanGelder et al 2017	<6h	2121	3,14	1,49 - 6,62	13.3*	NA
	≥6h	234			4.9*	
		24h				
Botto et al 2008	<24h	345	3,094	0,94 - 10,15	4	NA
	≥24h	223			8	
VanGelder et al 2017	<24h	2226	4,31	1,92 - 9,69	14.3*	NA
	≥24h	129			3.9*	
Kaplan et al 2019	<24h	24270	1,691	1,26 -2,28	206	NA
	≥24h	3762			54	

*** Corresponding events based on provided incidence rate**
IR: incidence rate

Table S3. Quality assessment of the selected studies based on the Newcastle-Ottawa Scale (NOS).

Study	Selection ☆☆☆☆/★★★★				Comparability ☆☆/★★	Outcome ☆☆☆/★★★			Overall stars	Quality Assessment
	Is the Case Definition Adequate?	Representativeness of the Cases	Selection of Controls	Definition of Controls	Comparability of Cases and Controls on the Basis of the Design or Analysis	Ascertainment of Exposure	Same method of ascertainment for cases and controls	Non-Response Rate		
Swiryn et al 2017	★	★	★	★	★★	★	★	★	9/9	Good quality
Study	**Selection ☆☆☆☆/★★★★**				**Comparability ☆☆/★★**	**Outcome ☆☆☆/★★★**			**Overall stars**	**Quality Assessment**
	Representativeness of the exposed cohort	**Selection of the non-exposed cohort**	**Ascertainment of exposure**	**Demonstration that outcome of interest was not present at the start of the study**	**Comparability of cohorts on the basis of the design or analysis controlled for confounders**	**Assessment of outcome**	**Was follow-up long enough for outcomes to occur**	**Adequacy of follow-up of cohorts**		
Botto et al 2008		★	★		★	★		★	5/9	Medium quality
VanGelder et al 2017	★	★	★	★	★★	★	★	★	9/9	Good quality
Perino et al 2019		★	★	★	★★	★	★		7/9	Good quality
Boriani et al 2014	★	★	★	★	★★		★		7/9	Good quality
Li et al 2019	★	★	★	★	★★	★	★		8/9	Good quality
Kim et al 2016	★	★	★	★	★★	★	★		8/9	Good quality
Jons et al 2011		★	★	★	★	★	★		6/9	Fair quality
Israel et al 2017		★	★	★			★	★	5/9	Medium quality
Gonzalez et al 2014	★	★	★	★	★★	★	★		8/9	Good quality
Glotzer et al 2003	★	★	★	★	★★	★	★		8/9	Good quality
Christensen et al 2014		★	★	★	★★	★		★	7/9	Good quality
Capucci et al 2005		★	★	★	★★		★		6/9	Fair quality
Amara et al 2017	★	★	★	★		★	★	★	7/9	Good quality
Kaplan et al 2019	★	★	★	★	★★	★	★	★	9/9	Good quality
Bertini et al 2010		★	★		★★	★	★		6/9	Fair quality
Caldwell et al 2009		★	★	★		★	★		5/9	Medium quality
Kawakamy et al 2017	★	★	★	★	★★	★	★		8/9	Good quality
Martin et al 2017	★	★	★	★	★★	★	★	★	9/9	Good quality
Miyazawa et al 2019	★	★	★	★	★★	★	★		8/9	Good quality
Nakano et al 2018	★	★	★	★	★★	★	★		8/9	Good quality
Ogino et al 2017	★	★	★	★	★★	★	★	★	9/9	Good quality
Pedersen et al 2018		★	★	★		★	★	★	6/9	Fair quality
Shanmugan et al 2011	★	★	★		★★	★	★		7/9	Good quality
Wilton et al 2016	★	★	★		★★	★	★	★	8/9	Good quality
Witt et al 2015	★	★	★	★	★★	★	★	★	9/9	Good quality
Healey et al 2012	★	★	★	★	★★	★	★		8/9	Good quality
Petrac et al 2012		★	★	★		★	★	★	6/9	Fair quality
Reiffel et al 2017		★	★	★		★	★		5/9	Medium quality

Table S4. Certainty assessment of the selected studies based on the GRADING system.

Number of studies	Study design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Other considerations	Certainty
Risk of stroke or systemic embolism based on the duration of longer AHRE							
30 seconds AHRE duration							
4	Observational and Randomized (1:1)	Few concerns	Low	Low	Moderate	Low publication Bias	Moderate
5 to 6minutes AHRE duration							
13	Observational	Low	Moderate	Low	Moderate	Low publication Bias	Moderate
24 hours AHRE duration							
5	Observational	Low	Moderate	Low	Low	Low publication Bias	High
Risk of stroke or systemic embolism based on the cumulative day-level AHRE burden							
24 hours AHRE duration							
2	Observational	Few concerns	Low	Low	Moderate		Moderate

Figure S1. Flow diagram of studies identified, screened and included in the meta-analysis.

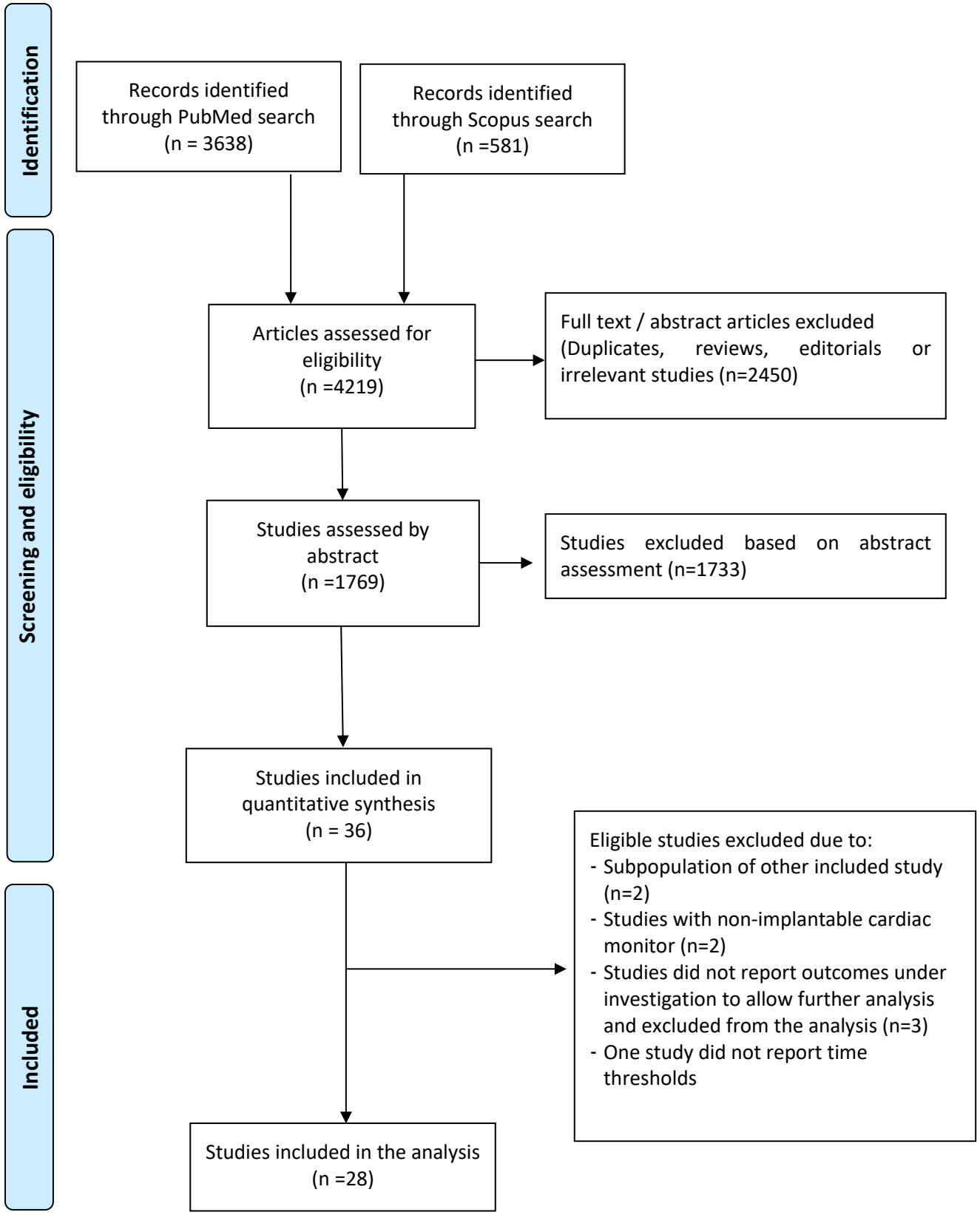
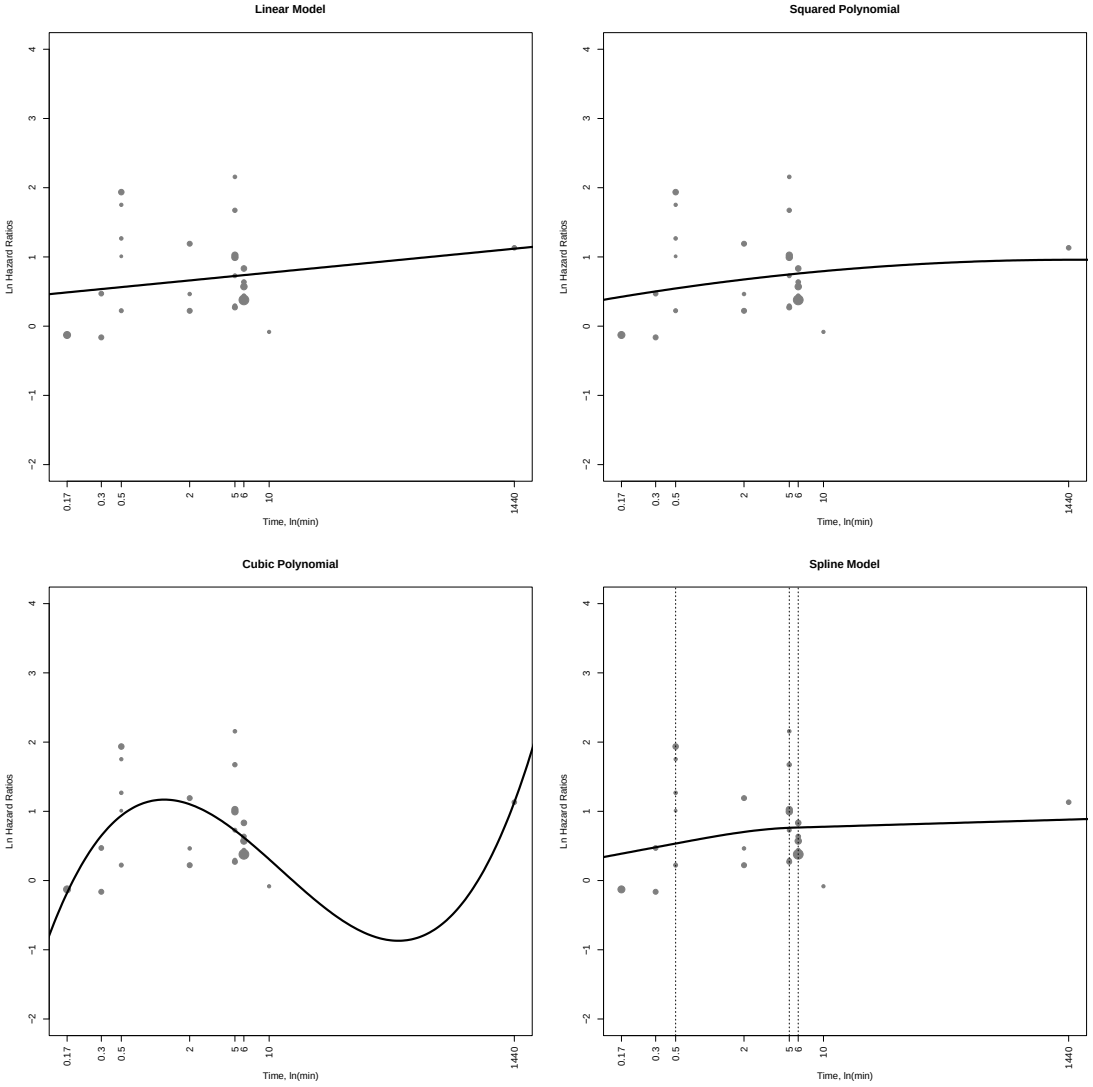


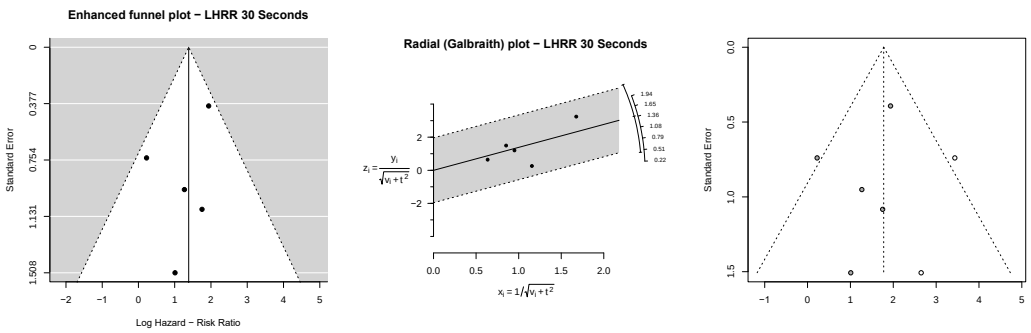
Figure S2. Linear and non-linear fixed effects meta regression based on the threshold of the longest AHRE duration.



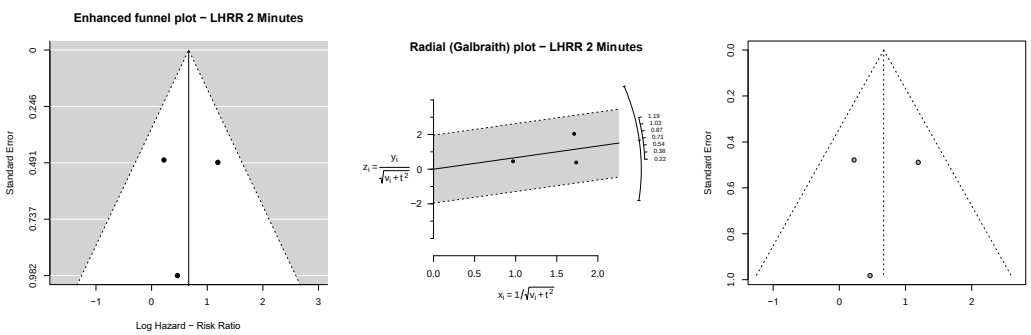
When the high-leverage study from Capucci et al 2005 was excluded there was still no significant association between increasing AHRE thresholds and the risk of stroke or systemic embolism (HR per 1 ln min increase=1.09, 95% CI 0.878-1.36, P=0.412).

Figure S3. Diagnostic plots for each time threshold.

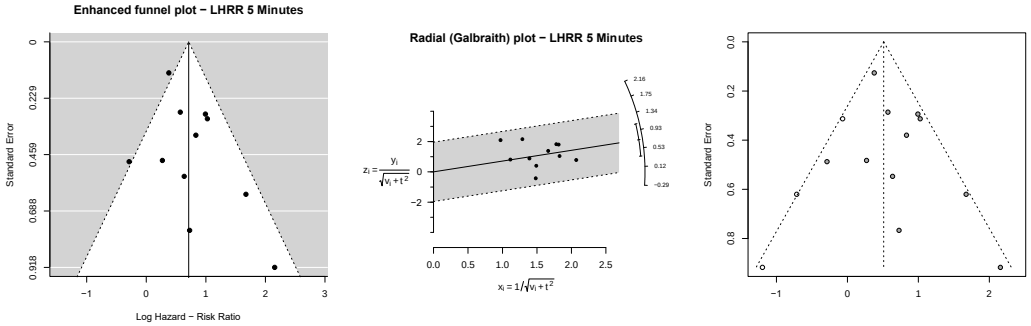
A



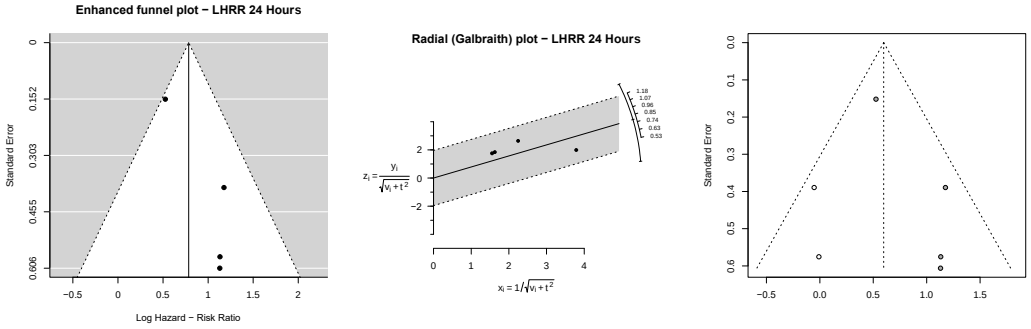
B



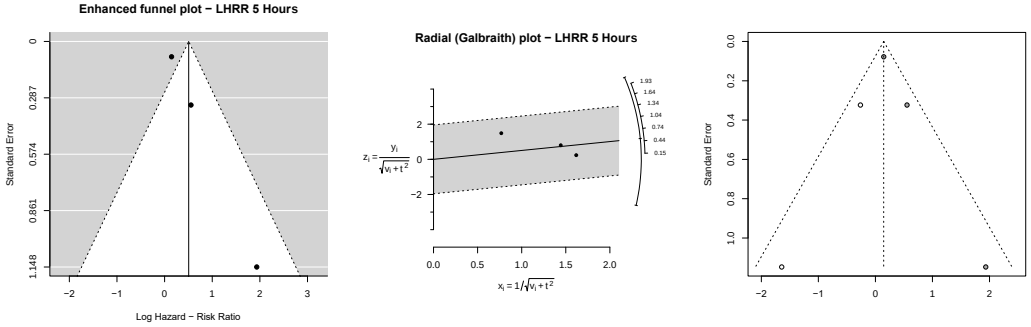
C



D



E



A: funnel plots and regression tests for the threshold of 30 seconds; B: funnel plots and regression tests for the threshold of AHRE >2 minutes; C: funnel plots and regression tests for the threshold of AHRE >5 minutes; D: funnel plots and regression tests for the threshold AHRE >24 hours; E: funnel plots and regression tests for the threshold cumulative day-level AHRE burden ≥24 hours